MT8002 Elements of Set Theory
7.2 Ordered Sets. Linearly Ordered Sets

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

Introduction
Binary Relations
Weak Orders
Strict Orders
Orders on Natural Numbers
Order-Isomorphism
Order-Embedding
Representation of Linearly Ordered Sets
Maximum and Minimum Elements
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Outline

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Introduction

- We can use binary relations to order some (partial order) or all (linear order) the elements of a set.
- Orderings are binary relations that satisfy certain properties.
- Let X be a set. Weak orders $(\leq_X)$ and strict orders $(<_X)$ can be interchangeably defined.

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Binary Relations

Definition (p. 164)

Let X be a set. A set R is a **binary relation on** X if $R \subseteq X \times X$.

Notation

Let R be a binary relation on a set X.

(a) We write $(x, y) \in R$ or xRy.

(b) We write $(x, y) \notin R$ or $x \not R y$.

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Weak Orders

Definitions (p. 164)

Let $\leq_X$ be a binary relation on a set X. The relation $\leq_X$ is a **weak partial order** on the set X (or the set X is **weakly partially ordered** by the relation $\leq_X$), if it has the following properties:

(a) **reflexive**: for all $x \in X$, $x \leq_X x$;

(b) **anti-symmetric**: for all $x, y \in X$, if $x \leq_X y$ and $y \leq_X x$ then $x \doteq y$;

(c) **transitive**: for all $x, y, z \in X$, if $x \leq_X y$ and $y \leq_X z$ then $x \leq_X z$.

Weak Orders

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(c) **transitive**: for all $x, y, z \in X$, if $x \leq_X y$ and $y \leq_X z$ then $x \leq_X z$.

If additionally the relation $\leq_X$ has the following property:

(d) **linear**: for all $x, y \in X$, $x \leq_X y$ or $y \leq_X x$,

then the relation $\leq_X$ is a **weak linear order** on the set X (or the set X is **weakly linearly ordered** by the relation $\leq_X$).

Weak Orders

Examples (informal, p. 164)

The usual relations $\leq$ and $\geq$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are weak linear orders on those sets.

Weak Orders

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Example

Let X be a set. The relation $\subseteq$ on $\mathcal{P}(X)$ is a weak partial order.

Weak Orders

Exercise

Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Weak Orders

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Let X be a set. Is the relation $\subseteq$ on $\mathcal{P}(X)$ a weak linear order?

Answer

No! For example the relation $\subseteq$ on the power set of $\{a, b\}$ is no linear because $\{a\}, \{b\} \in \mathcal{P}(\{a, b\})$ but neither $\{a\} \nsubseteq \{b\}$ not $\{b\} \nsubseteq \{a\}$.

Weak Orders

Description

The **restriction** of a weak partial order to a subset is the weak partial order obtained by discarding all comparisons involving elements outside of the subset.

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Definition (p. 164)

Let R be a weak partial order on a set X and let $A \subseteq X$. The **restriction** of the relation R to the set A, denoted $R|_{A\times A}$, is the relation defined by

$$R|_{A\times A} := \{ (a,b) \in R : a,b \in A \}.$$

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$$R|_{A \times A} := \{ (a, b) \in R : a, b \in A \} .$$

Theorem (Exercise 7.2)

Let R be a weak partial (respectively linear) order on a set X and let $A \subseteq X$. Show that $R|_{A \times A}$ is a partial (respectively linear) order on the set A.

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Strict Orders

Definitions (p. 165)

Let $<_X$ be a binary relation on a set X. The relation $<_X$ is a **strict partial order** on X (or the set X is **strictly partially ordered** by the relation $<_X$) if it has the following properties:

(a) **irreflexive**: for all $x \in X$, it is not the case that $x <_X x$;

(b) **transitive**: for all $x, y, z \in X$, if $x <_X y$ and $y <_X z$ then $x <_X z$.

Strict Orders

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If additionally the relation $<_X$ has the following property:

(c) **linear**: for all $x, y \in X$, $x <_X y$ or $x \doteq y$ or $y <_X x$,

then the relation $<_X$ is a **strict linear order** on the set X (or the set X is **strictly linearly ordered** by the relation $\leq_X$).

Strict Orders

Examples

The usual relations $<$ and $>$ on the sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ are strict linear orders on those sets.

Example

Let X be a set. The relation $\subset$ on $\mathcal{P}(X)$ is a strict partial order.

Strict Orders

Theorem (relationship between weak and strict partial orders, Exercise 7.4)

(a) Let $\leq_X$ be a weak partial order on a set X. The relation

$$x <_X y := x \leq_X y \text{ and } x \doteq\!\!\!\!/\; y$$

is a strict partial order on the set X.

Strict Orders

Theorem (relationship between weak and strict partial orders, Exercise 7.4)

(a) Let $\leq_X$ be a weak partial order on a set X. The relation

$$x <_X y := x \leq_X y \text{ and } x \not\doteq y$$

is a strict partial order on the set X.

(b) Let $<_X$ be a strict partial order on a set X. The relation

$$x \leq_X y := x <_X y \text{ or } x \doteq y$$

is a weak partial order on the set X.

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An "extra" property of natural numbers

$$\mathbf{0} \in \mathbf{1} \in \mathbf{2} \in \mathbf{3} \in \cdots$$

$$\mathbf{0} \in \{\mathbf{0}\} \in \{\mathbf{0}, \mathbf{1}\} \in \{\mathbf{0}, \mathbf{1}, \mathbf{2}\} \in \cdots$$

$$\emptyset \in \{\emptyset\} \in \{\emptyset, \{\emptyset\}\} \in \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \in \cdots$$

Orders on Natural Numbers

Theorem 3.6

The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the relation $\in$.

Orders on Natural Numbers

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The set of natural numbers $\mathbb{N}$ is strictly linearly ordered by the relation $\in$.

That is, the relation $\in$ on $\mathbb{N}$ has the following properties:

(a) **irreflexive**: for all $\mathbf{n} \in \mathbb{N}$, $\mathbf{n} \notin \mathbf{n}$;

(b) **transitive**: for all $\mathbf{m}, \mathbf{n}, \mathbf{p} \in \mathbb{N}$, if $\mathbf{m} \in \mathbf{n}$ and $\mathbf{n} \in \mathbf{p}$ then $\mathbf{m} \in \mathbf{p}$;

(c) **linear**: for all $\mathbf{m}, \mathbf{n} \in \mathbb{N}$, $\mathbf{m} \in \mathbf{n}$ or $\mathbf{m} \doteq \mathbf{n}$ or $\mathbf{n} \in \mathbf{m}$.

Orders on Natural Numbers

Example (p. 42)

Although the relation $\in$ is transitive on the set ω it does not mean that the relation is transitive on any set. For example, the relation is not transitive on the set

$$\{\mathbf{0}, \{\mathbf{0}\}, \{\{\mathbf{0}\}\}\}$$

because $\mathbf{0} \in \{\mathbf{0}\}$ and $\{\mathbf{0}\} \in \{\{\mathbf{0}\}\}$ but $\mathbf{0} \notin \{\{\mathbf{0}\}\}$.

Orders on Natural Numbers

Definition (p. 41)

For all $\mathbf{m}, \mathbf{n} \in \mathbb{N}$,

$\mathbf{m} < \mathbf{n} := \mathbf{m} \in \mathbf{n}$,
$\mathbf{m} \leq \mathbf{n} := \mathbf{m} < \mathbf{n}$ or $\mathbf{m} \doteq \mathbf{n}$.

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Order-Isomorphism

Convention

From now on a (weakly or strictly; partially or linearly) ordered set X by a relation R will we denote by the pair (X, R).[†]

[†]Note that the textbook does not use this convention.

Order-Isomorphism

Convention

From now on a (weakly or strictly; partially or linearly) ordered set X by a relation R will we denote by the pair (X, R).[†]

Description

We need a precise way to express when two ordered sets are "the same".

[†]Note that the textbook does not use this convention.

Order-Isomorphism

Definitions (p. 166)

Let $(X, <_X)$ and $(Y, <_Y)$ be two strictly partially ordered sets.

These sets are **order-isomorphic**, denoted $(X, <_X) \cong (Y, <_Y)$, if there is a bijection $f : X \longrightarrow Y$ such that for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

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The map f is an **order-isomorphism**.

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The map f is an **order-isomorphism**.

Remark

The above definition on weakly partially ordered sets is similar.

Order-Isomorphism

Exercise (informal)

Let $<$ and $>$ be the usual orders on $\mathbb{Z}$. Show that $(\mathbb{Z}, <) \cong (\mathbb{Z}, >)$.

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Let $<$ and $>$ be the usual orders on $\mathbb{Z}$. Show that $(\mathbb{Z},<) \cong (\mathbb{Z},>)$.

Solution

Let $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be the function $n \longmapsto -n$. We can show that the function f is a bijection and that it preserves the order, that is, for all $m, n \in \mathbb{Z}$,

$$m < n \quad \text{if and only if} \quad -m > -n.$$

Order-Isomorphism

Theorem (Exercise 7.6)

The relation $\cong$ is an equivalence relation on strictly partially ordered sets, i.e., the relation $\cong$ satisfies the following properties:

(a) **reflexive**: for all X, $X \cong X$;

(b) **symmetric**: for all X and Y, if $X \cong Y$ then $Y \cong X$;

(c) **transitive**: for all X, Y and Z, if $X \cong Y$ and $Y \cong Z$ then $X \cong Z$.

Order-Isomorphism

Definition (p. 168)

An **order-theoretic invariant** is a property of an ordered set preserved under order-isomorphism.

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Order-Embedding

Description

We also need to define when an ordered set fits inside another while preserving its order structure entirely.

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Definition (p. 167)

Let $(X, <_X)$ and $(Y, <_Y)$ be two strictly partially ordered sets.

A function $f : X \longrightarrow Y$ is an **order-embedding** of $(X, <_X)$ into $(Y, <_Y)$ if f is one-one and, for all $x_1, x_2 \in X$,

$$x_1 <_X x_2 \quad \text{if and only if} \quad f(x_1) <_Y f(x_2).$$

Order-Embedding

Exercise (informal)

Let $<$ be the usual order on $\mathbb{N}$ and $\mathbb{R}$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

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Let $<$ be the usual order on $\mathbb{N}$ and $\mathbb{R}$. Show an order-embedding of $(\mathbb{N}, <)$ into $(\mathbb{R}, <)$

Solution

Let $f : \mathbb{N} \longrightarrow \mathbb{R}$ be the function $n \longmapsto n$. We can show that the function f is one-one and that it preserves the order, that is, for all $m, n \in \mathbb{N}$,

$$m > n \quad \text{if and only if} \quad f(m) > f(m).$$

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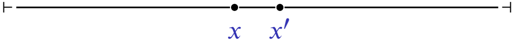
Representation of Linearly Ordered Sets

"*If $<_x$ is the order on X, and the elements x, x' of X are related by $x <_x x'$, we picture x to the left of x'. [...] Of course, we must not read too much into the picture. The picture is very suggestive of the usual real line, but for instance there might not be any elements of X between x and x' in the picture above; and indeed there might be no order-embedding from X into $\mathbb{R}$ with its usual order.*" *(p. 167)*



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Maximum and Minimum Elements

Definitions (p. 168)

Let $(X, \leq_X)$ be a weakly linearly ordered sets and let $a, b \in X$.

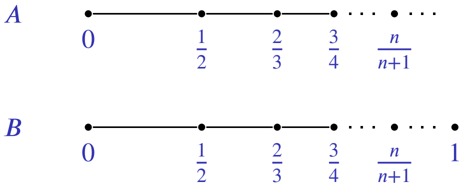
(a) The element b is the **maximum** (or **greatest**) element of X, if $x \leq_X b$ for all $x \in X$.

(b) The element a is the **minimum** (or **least**) element of X, if $a \leq_X x$ for all $x \in X$.

Maximum and Minimum Elements

Example (informal, p. 168)

Let $A = \left\{ 1 - \frac{1}{n+1} \ : n \in \mathbb{N} \right\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.



The set A has minimum element, 0, but no maximum element.
The set B has minimum element, 0, and maximum element, 1.

Maximum and Minimum Elements

Remark

In the next theorem we shall see that the maximum and the minimum are order-theoretic invariants.

Maximum and Minimum Elements

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Theorem 7.1

Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two weakly linearly ordered sets.

Let $f : X \longrightarrow Y$ be an order-isomorphism from $(X, \leq_X)$ to $(Y, \leq_Y)$.

(a) If X has a maximum element then so does Y.

(b) If X has a minimum element then so does Y.

Maximum and Minimum Elements

Example (informal)

Let $A = \left\{ 1 - \frac{1}{n+1} : n \in \mathbb{N} \right\}$ and $B = A \cup \{1\}$, subsets of $\mathbb{Q}$, with the usual order $\leq$.

From Theorem 7.1 the ordered sets $(A, \leq)$ and $(B, \leq)$ are not order-isomorphic because the set B has maximum element, but the set A does not.

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Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).